

MID-TERM EXAMINATION

Course: **COMPUTER SYSTEM PROGRAMMING**

Time: **60 minutes**

Term: 1 – Academic year: **2021-2022**

Lecturer(s): **Dinh Dien**

Student name:

Student ID:

Problem 1 (20 point).

Assume we are running code on a 5-bit binary and 2-digit hexadecimal machine and we can represent a decimal number in 1's complement, 2's complement and bias 10. Fill in the empty boxes. Write number in 2-digit hexadecimal format. If the number can not be represented, please fill “_”.

Description	1's complement	2's complement	Bias 10
0			
-9			
13			
-15			
23			

Short explanation: _____

Problem 2 (20 point).

Suppose we have 7-bit computer that uses IEEE floating point arithmetic where floating point number has 1 sign bit, 3 exponent bits, and 3 fraction bits. All of the bits in the hardware work properly.

For each of the following, write the binary value and the corresponding decimal value of the 7-bit floating point number that is the closet available representation of the requested number. If rounding is necessary use round-to-nearest. Give the decimal value either as whole number or fractions. The first line is filled in for you.

Number	Binary	Decimal
0	0 000 000	0.0
0.125		
Smallest positive normalized number		
Smallest positive denormalized number > 0		
Largest positive number < ∞		
3.1		

12.25

Short explanation: _____

Problem 3 (20 point).

In the following questions assume the variables a and b are signed integers and that the machine uses two's complement representation. Also assume that MAX_INT is the maximum integer, MIN_INT is the minimum integer, and W is one less than the word length (e.g., W=31 for 32-bit integers).

Match each of the descriptions on the left with a line of code on the right (write in the letter).

1. a&b	a. $\sim(\sim a \mid (b \wedge (\text{MIN_INT} + \text{MAX_INT})))$
2. a*22	b. $((a \wedge b) \& \sim b) \mid (\sim(a \wedge b) \& b)$
3. a	c. $1 + (a \ll 3) + \sim a$
4. !a	d. $(a \ll 4) + (a \ll 2) + (a \ll 1)$
5. (a<0)?1:-1	e. $a \wedge (\text{MIN_INT} + \text{MAX_INT})$
	f. $\sim((a \mid (\sim a + 1)) \gg W) \& 1$
	g. $\sim((a \gg W) \ll 1)$
	h. $a \gg 2$

Short explanation: _____

Problem 4 (20 point).

Address	Value	Register	Value
0x100	0xFF	%eax	0x100
0x104	0xAB	%ecx	0x1
0x108	0x09	%edx	0x3
0x10C	0x11		

Fill in the following table the effect of the following instructions, both in term of the register or memory location that will be updated and the resulting value:

Instruction	Destination	Value
subl %edx,4(%eax)		
addl 8(%eax),%edx		
addl %edx,%eax		
imull \$2,-16(%eax,%edx)		
incl (%eax)		

Short explanation: _____

Problem 5. (20 point)

Complete the C code

<pre>/* ptr=%rdi, x=%esi */ int f2(int *ptr, int x) { int i = ____; while (____ 0) { ____ = x; ____; } return i; }</pre>	<pre>f2: movl \$0,%eax .L2: movslq %eax,%rdx leaq (%rdi,%rdx,4),%rdx cmpl \$0, (%rdx) jg .L4 movl %esi, (%rdx) addl \$1,%eax jmp .L2 .L4: ret</pre>
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Short explanation: _____